

ECON 605: Macroeconomic Theory II

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Office Hours: Weds 1:00-2:30pm

Office: KAP 312

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Lecture: Tues/Thurs 10:00-11:50am, KAP 134

TA: Xuemeng Yin xuemengy@usc.edu

Course Description

ECON 605 is the second semester (third quarter) of the macroeconomics sequence for first-year Ph.D. students at USC. We will spend most of this class discussing modern business cycle theory and dynamic stochastic general equilibrium (DSGE) models.

We start by developing a theory of real business cycles (RBCs) by extending the Ramsey-Cass-Koopmans neoclassical growth model to incorporate stochastic processes and endogenous labor supply. We evaluate the claim that these models can replicate key empirical patterns describing business cycles. We pay particular attention to the analytical and numerical toolkit that we need to work with these dynamic general equilibrium models: equilibrium concepts and properties; steady states; transition dynamics; linearization; impulse responses and variance decompositions; calibration and Bayesian estimation.

In the second half of the course, we further extend this real business cycle framework to incorporate imperfectly competitive markets, nominal rigidities, and monetary policy. We use this New Keynesian framework to study the transmission of fiscal and monetary policy to aggregate economic activity, and discuss the successes and failures of this approach in detail. In the final part of this course, we discuss extensions of the New Keynesian framework that are commonly encountered in modern macroeconomics, including labor market frictions, financial market frictions, and heterogeneous agents.

Prerequisites

Students are expected to have successfully completed ECON 601 (Microeconomic Theory I) and ECON 602 (Macroeconomic Theory I).

Required and Recommended Materials

There are *no required textbooks*. Lecture slides, lecture notes, and problem sets are the primary materials for this course, and are intended to be self-contained to the greatest extent possible.

Where appropriate, I include references at the back of lecture slides with additional material (e.g. papers, notes, slides) from other sources that you may find useful for study purposes.

The following textbooks are **optional**, but potentially useful, references for our course material:

1. *Recursive Macroeconomic Theory*, by Lars Ljungqvist and Thomas Sargent
2. *Interest and Prices*, by Michael Woodford
3. *Monetary Policy, Inflation, and the Business Cycle*, by Jordi Galí

Course Structure

Lecture. Lectures are held twice a week. Please attend if you can.

Problem Sets. There are four problem sets for this quarter. Assignment dates and due dates are listed under the Schedule section of the syllabus. Problem sets are primarily a study aid and commitment device for you to get hands-on experience with the material before the exam. Problem sets are graded entirely on completion. You can (and should) work with your peers on the problem sets; however, every student needs to submit their own written up solutions.

Sections and TA. We will hold sections led by our amazing teaching assistant, Xuemeng Yin. We will have more details on sections to share in the second week of class.

Final Exams. We will have two exams in the last week of class. I will post two practice exams one week before the first of these exams. Please let me know *well in advance* if you have conflicts that prohibit you from attending one or both of these exams and we can discuss a make-up exam at an alternative date.

The first exam (March 10) will cover approximately the first half of our course material on real business cycles. The second exam day (March 12) will cover the second half of our material on New Keynesian models. These exams are held together at the end of class to try to make them somewhat more “fresh” for your core exams in May.

Grading

80% of your grade is determined by our two in-class exams in the final week of class (40% each). 20% of your grade is determined by completing four problem sets.

Course Policies

During Class

Electronic devices are allowed if they are used for note-taking. Please do not be disruptive to me or your classmates during lecture.

Large Language Models

You are free to use large language models (LLMs) as a study aid, just as you are free to use any resource online. Note that these tools (like any other resource) can sometimes provide you with incorrect information, and they can also tempt you away from learning the material. You **will not** be able to use LLMs (or other electronic tools) during our in-class exams, so it is important that if you choose to use these tools, you use them as a device to help learn the material rather than as a device to avoid learning the material.

Attendance

I do not track class attendance. You do not need to inform me about absences, and you should not attend class if you are feeling sick. You should attend class if you can..

Late Assignments

Please let me know in advance if you don't think you can turn in a problem set before the designated due date. Problem sets are posted according to the schedule on this syllabus. Problem sets are due *before class* (10:00AM) on the designated due date. Incomplete or late work receives a zero.

Academic Integrity and Honesty

The University of Southern California is foremost a learning community committed to fostering successful scholars and researchers dedicated to the pursuit of knowledge and the transmission of ideas. Academic misconduct is contrary to this fundamental mission and includes any act of dishonesty in the submission of academic work (either in draft or final form), as well as cheating, plagiarism, fabrication (e.g., falsifying data), knowingly assisting others in acts of academic dishonesty, and any act that gains or is intended to gain an unfair academic advantage. Students are expected to uphold the highest standards of academic integrity in all coursework.

This course follows the expectations for academic integrity as stated in the [USC Student Handbook](#). All students are expected to submit assignments that are original work and prepared specifically for the course/section in this academic term. Students may not submit work written by others or recycle work prepared for other courses without obtaining written permission from the instructor(s). Students suspected of academic misconduct will be reported to the Office of Academic Integrity.

Academic dishonesty has a far-reaching impact and is considered a serious offense against the university. Violations will result in a grade penalty, such as a failing grade on the assignment or in the course, and disciplinary action from the university, such as suspension or expulsion.

For more information about academic integrity see the [Student Handbook](#), the [Office of Academic Integrity's website](#), and university policies on [Research and Scholarship Misconduct](#).

Please ask me if you are unsure what constitutes unauthorized assistance on an exam or assignment or what information requires citation and/or attribution.

Schedule

The following table describes the topics and dates of each lecture, as well as our final exams and problem sets.

This syllabus and schedule is a **living document** and may be updated as the quarter progresses. Please make sure you are looking at the most recent version of the syllabus posted online.

Date	Topic	Assignments
January 13	A Brief History of Business Cycle Theory	
January 15	Real Business Cycles	
January 20	Competitive Equilibrium and the Welfare Theorems	Problem set 1 posted
January 22	Shocks and Propagation in Real Business Cycles	
January 27	Evaluating Real Business Cycle Theory	Problem set 1 due
January 29	Money, Monetary Neutrality and the Classical Dichotomy	Problem set 2 posted
February 03	Imperfect Competition, Nominal Rigidities, and Real Rigidities	
February 05	The New Keynesian Model	Problem set 2 due
February 10	Properties of the New Keynesian Model	Problem set 3 posted
February 12	Monetary Economics, According to New Keynesians	
February 17	Fiscal Policy in the New Keynesian Model	Problem set 3 due
February 19	Liquidity Traps and the Zero Lower Bound	Problem set 4 posted
February 24	"Puzzles," According to New Keynesians	
February 26	Labor Market Frictions	Problem set 4 due
March 3	Open Economies	Review posted
March 5	Heterogeneity and Incomplete Markets	
March 10	In-Class Exam Part I (RBC)	
March 12	In-Class Exam Part II (NK)	

Statement on University Academic and Support Systems

Students and Disability Accommodations:

USC welcomes students with disabilities into all of the University's educational programs. The Office of Student Accessibility Services (OSAS) is responsible for the determination of appropriate accommodations for students who encounter disability-related barriers. Once a student has completed the OSAS process (registration, initial appointment, and submitted documentation) and accommodations are determined to be reasonable and appropriate, a Letter of Accommodation (LOA) will be available to generate for each course. The LOA must be given to each course instructor by the student and followed up with a discussion. This should be done as early in the semester as possible as accommodations are not retroactive. More information can be found at osas.usc.edu. You may contact OSAS at (213) 740-0776 or via email at osasfrontdesk@usc.edu.

Student Financial Aid and Satisfactory Academic Progress:

To be eligible for certain kinds of financial aid, students are required to maintain Satisfactory Academic Progress (SAP) toward their degree objectives. Visit the Financial Aid Office webpage for undergraduate- and graduate-level SAP eligibility requirements and the appeals process.

Support Systems:

Counseling and Mental Health - (213) 740-9355 24/7 on call

Free and confidential mental health treatment for students, including short-term psychotherapy, group counseling, stress fitness workshops, and crisis intervention.

988 Suicide and Crisis Lifeline - 988 for both calls and text messages 24/7 on call

The 988 Suicide and Crisis Lifeline (formerly known as the National Suicide Prevention Lifeline) provides free and confidential emotional support to people in suicidal crisis or emotional distress 24 hours a day, 7 days a week, across the United States. The Lifeline consists of a national network of over 200 local crisis centers, combining custom local care and resources with national standards and best practices. The new, shorter phone number makes it easier for people to remember and access mental health crisis services (though the previous 1 (800) 273-8255 number will continue to function indefinitely) and represents a continued commitment to those in crisis.

CARE-SC: Confidential Advocacy, Resources, and Education Support Center - (213) 740-9355

Confidential advocates, prevention educators, and professional counseling teams work to promote a universal culture of consent, as well as prevent and respond to sexual assault, intimate partner violence, stalking, or other relationship harm. Services available to all USC students at no cost.

Office of Civil Rights Compliance - (213) 740-5086

Information about how to get help or help someone affected by harassment, discrimination, retaliation on the basis of a protected characteristic, rights of protected classes, reporting options, and additional resources for students, faculty, staff, visitors, and applicants.

USC Report & Response - (213) 740-2500

The USC Report & Response website is the university's central reporting portal for concerns arising in the academic space or workplace. All concerns will be assessed and referred to the appropriate university office for resolution. Any questions about USC Report & Response or reporting, in general, can be referred to the Office of Professionalism and Ethics at ope@usc.edu.

USC Campus Support and Intervention - (213) 740-0411

Focuses on student success by assisting students in navigating and resolving complex issues through problem solving, presenting options, and connecting to resources.

USC Emergency Information

Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible.

USC Department of Public Safety

For 24 hour emergency assistance or to report a crime: UPC: (213) 740-4321, HSC: (323)-442-1000.

For 24 hour non-emergency assistance or information: UPC: (213) 740-6000, HSC: 323-442-1200.

Office of the Ombuds - (213) 821-9556 (UPC) / (323-442-0382 (HSC)

A safe and confidential place to share your USC-related issues with a University Ombuds who will work with you to explore options or paths to manage your concern.

Occupational Therapy Faculty Practice - (323) 442-2850 or otfp@med.usc.edu

Confidential Lifestyle Redesign services for USC students to support health promoting habits and routines that enhance quality of life and academic performance.